

EPT-TEST-16(IIIT Hyderabad Full Length Test - 3)

Question Paper Contain

50 Aptitude | 6 Algorithms | 9 C-Programming & DS |

11 Maths | 6 CN | 6 DBMS | 8 OS | Digital 4 |

Total 100 Marks | 180 Minutes Duration

+1 Mark to each correct answer

-0.25 Mark to each wrong answer

1.) The _____ is excessively elevated to be deemed _____.

- A. fair / fare
- B. faer / fair
- C. fare / fare
- D. fare / fair

2.) If r is a root of the quadratic equation $x^2+2x+6=0$, what is the value of the expression $(r+2)(r+3)(r+4)(r+5)$?"

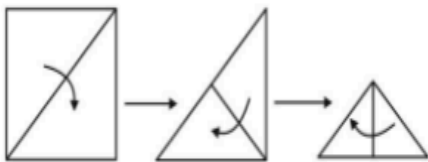
- A. 126
- B. 125
- C. -126
- D. -125

3.) find the last digit of $(100008)^{12499}$?

- A. 8
- B. 2
- C. 4
- D. 6

4.) As shown in the figure below, a square piece of paper is folded three times along its diagonal to get an isosceles triangle whose equal sides are 10 cm long

The area of the unfolded original piece of paper in sq.cm. is _____



- A. 400
- B. 300
- C. 200
- D. 100

5.) Consider two strings $S1 = \text{"ROHITSHARMA"}$ and $S2 = \text{"HARDIKPANDYA"}$. Let x be the length of a longest common subsequence between $S1$ and $S2$ and let y be the number of such unique longest common subsequences of length x between $S1$ and $S2$. What is $x + 5y$?

- A. 16
- B. 19
- C. 15
- D. 16

6.) What must be added to each term of the ratio 2:5 so that it may equal to 5:6?

- A. 65
- B. 78
- C. 13
- D. 12

7.) Find the next term in the series

10, 19, 30, 43, 58?

- A. 75
- B. 65
- C. 70
- D. 76

8.) Choose the word which is different from the rest.

- A. Cap
- B. Turban
- C. Helmet
- D. Veil

9.) How many trailing zeros will be there after the rightmost non-zero digit in the value of $25!$

- A. 6
- B. 5
- C. 7
- D. 8

10.) *How many numbers between 0 and 1 million can be formed using 0,7,8 (numbers greater than 0 and less than 1 million)*

- A. 700
- B. 750
- C. 728
- D. 738

11.) there are 2 Indians and 4 Chinese in a group of 6 people. how many subgroups of this group can we choose so that every subgroup has at least one Indian?

- A. 40
- B. 48
- C. 56
- D. 64

12.) If $(7 \text{ a } 13) \text{ b } 3 = 60$, and $(12 \text{ a } 13) \text{ b } 4 = 100$, then $(30 \text{ a } 31) \text{ b } 3 = ?$

- A. 313
- B. 219
- C. 343
- D. 183

13.) What is the area of a rectangle with the largest perimeter that can be inscribed in the circle (i.e., all the vertices of the rectangle are on the circle with radius 2)?

- A. 1
- B. 8
- C. 3
- D. 4

14.) Synonym: PHLEGMATIC

- A. practical
- B. salivary
- C. dishonest
- D. calm

15.) Which of the above sentences are grammatically CORRECT?

- I. Rohit and HARDIK are here.
 - II. Rohit and Hardik is here.
 - III. Rohit's families is here.
 - IV. Rohit's family is here.
- A. (i) and (ii)
 - B. (i) and (iv)
 - C. (ii) and (iv)
 - D. (iii) and (iv)

16.) x and y are two operators on numbers p and q such that $p y q = p - q$, and $p x q = p \times q$

Then, $(9y(6x7)) - y(7x(6y4)) =$

- A. 40
- B. -26
- C. -33
- D. -47

17.) Four persons W, X, Y and Z are to be seated in a row. Y should not be seated at the second position from the left end of the row.

The number of distinct seating arrangements possible is:

- A. 6
- B. 9
- C. 18
- D. 24

18.) The author said, "Musicians rehearse before their concerts. Actors rehearse their roles before the opening of a new play. On the other hand, I find it strange that many public speakers think they can just walk onto the stage and start speaking. In my opinion, it is no less important for public speaker to rehearse their talks."

Based on the above passage, which one of the following is TRUE?

- A. The author is of the opinion that rehearsing is important for musicians, actors and public speakers
- B. The author is of the opinion that rehearsing is less important for public speakers than for musicians and actors
- C. The author is of the opinion that rehearsing is more important only for musicians than public speakers
- D. The author is of the opinion that rehearsal is more important for actors than musicians

19.) Five persons P, Q, R, S and T are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and T cannot be seated at either end of the row. P should

not be seated adjacent to S. R is to be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is:

- A. 2
- B. 3
- C. 4
- D. 5

20.) Consider the following sentences:

- i. The number of candidates who appear for the GATE examination is staggering.
- ii. A number of candidates from my class are appearing for the GATE examination.
- iii. The number of candidates who appear for the GATE examination are staggering.
- iv. A number of candidates from my class is appearing for the GATE examination.

Which of the above sentences are grammatically CORRECT?

- A. (i) and (ii)
- B. (i) and (iii)
- C. (ii) and (iii)
- D. (ii) and (iv)

21.) A digital watch X beeps every 40 seconds while watch Y beeps every 42 seconds. They beeped together at 10 AM.

The immediate next time that they will beep together is _____

- A. 10.08 AM
- B. 10.14 AM
- C. 11.00 AM
- D. 10.00 PM

22.) The front door of Ms. Sudo's house faces East. Ms Sudo leaves the house, walking 50 m straight from the back door that is situated directly opposite to the front door. She then turns to her right,

walks for another 100 m and stops. The direction of the point Ms. Sudo is now located at with respect to the starting point is _____

- A. South-East
- B. North-East
- C. West
- D. North-West

23.) Given below are two statements 1 and 2, and two conclusions I and II.

- Statement 1: All entrepreneurs are wealthy.
- Statement 2: All wealthy are risk seekers.
- Conclusion I: All risk seekers are wealthy.
- Conclusion II: Only some entrepreneurs are risk seekers.

Based on the above statements and conclusions, which one of the following options is CORRECT?

- A. Only conclusion I is correct
- B. Only conclusion II is correct
- C. Neither conclusion I nor II is correct
- D. Both conclusions I and II are correct

24.) A box contains 15 blue balls and 45 black balls. If 2 balls are selected randomly, without replacement, the probability of an outcome in which the first selected is a blue ball and the second selected is a black ball, is _____

- A. $\frac{3}{16}$
- B. $\frac{45}{236}$
- C. $\frac{1}{4}$
- D. $\frac{3}{4}$

25.) The world is going through the worst pandemic in the past hundred years.

The air travel industry is facing a crisis, as the resulting quarantine requirement for travelers led to weak demand.

In relation to the first sentence above, what does the second sentence do?

- A. Restates an idea from the first sentence
- B. Second sentence entirely contradicts the first sentence
- C. The two statements are unrelated
- D. States an effect of the first sentence

26.) Consider the following sentences:

- i. After his surgery, Raja hardly could walk.
- ii. After his surgery, Raja could barely walk.
- iii. After his surgery, Raja barely could walk.
- iv. After his surgery, Raja could hardly walk.

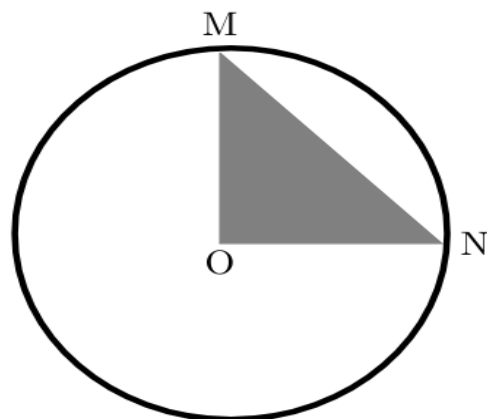
Which of the above sentences are grammatically CORRECT?

- A. (i) and (ii)
- B. (i) and (iii)
- C. (iii) and (iv)
- D. (ii) and (iv)

27.)

In the above figure, O is the centre of the circle and, M and N lie on the circle. The area of the right triangle MON is 50cm^2 . What is the area of the circle in cm^2 ?

- A. 2π
- B. 50π
- C. 75π
- D. 100π



28.)

If {

" $\oplus$ " means "-"

" $\otimes$ " means " $\div$ "

" Δ " means "+"

" ∇ " means "x"

then, the value of the expression $\Delta 3 \oplus 3 \Delta ((4 \otimes 2) \nabla 4) =$

- A. -1
- B. 8
- C. 7
- D. 6

29.) "The recent rise in the consumption of leafy vegetables suggests a shift towards healthier habits among the state's residents."

Which of the following conclusions can be logically drawn from the statement above?

- A. Prior to this increase, the state's residents rarely consumed leafy vegetables.
- B. Consuming leafy vegetables might not solely determine a healthy lifestyle.
- C. A diet rich in leafy vegetables is associated with leading a healthier life.
- D. The state's residents are becoming more conscious of the health risks associated with junk food consumption.

30.) In the wild, oxpeckers and rhinos engage in a mutualistic relationship. Oxpeckers alert rhinos to the presence of potential

threats like poachers, potentially safeguarding the rhinos.
Additionally, oxpeckers consume ticks that parasitize rhinos.

In the described mutualistic relationship, the primary advantages for oxpeckers and rhinos are, respectively,

A. Oxpeckers obtain sustenance, while rhinos derive no benefits.

B. Oxpeckers shield their environment from poachers, with rhinos receiving no advantages.

C. Oxpeckers secure a food supply, and rhinos could potentially be protected from poachers.

D. Oxpeckers prolong the lives of poachers, and rhinos ensure their own survival.

31.) In farm A, there are 65 hens, 91 ducks, and 169 goats. Farm B, located nearby, houses a total of 416 animals, comprising hens, ducks, and goats. The distribution ratio of hens to ducks to goats in farm B is 5:14:13. All animals from farm B are transferred to farm A. What is the revised ratio of hens to ducks to goats in farm A after the transfer?

A. 5:7:13

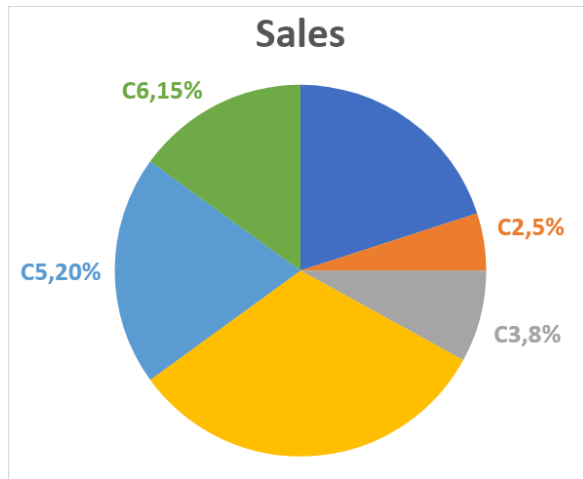
B. 5:14:13

C. 10:21:26

D. 21:10:26

COMPANY	ratio
---------	-------

32.)



The distribution of employees at the rank of executives, across different companies $C_1, C_2, \dots, C_6$ is presented in the chart given above. The ratio of executives with a management degree to those without a management degree in each of these companies is provided in the table above. The total number of executives across all companies is 10,000.

The total number of management degree holders among the executives in companies C_2 and C_5 together is _____.

- A. 225
- B. 600
- C. 1900
- D. 2500

33.) Five persons P, Q, R, S and T are sitting in a row not necessarily in the same order. Q and R are separated by one person, and S should not be seated adjacent to Q. The number of distinct seating arrangements possible is:

- A. 4
- B. 8
- C. 10
- D. 16

34.)

Getting to the top is _____ than staying on top.

- A. more easy
- B. much easy
- C. easiest
- D. easier

35.) $\oplus$ and $\odot$ are two operators on numbers p and q such that $p \oplus q = (p^2 + q^2)/pq$ and $p \odot q = p^2/q$

If $x \oplus y = 2 \odot 2$, then $x =$

- A. $y/2$
- B. y
- C. $3y/2$
- D. $2y$

36.) Four persons P, Q, R and S are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and R cannot sit adjacent to each other. S should be seated to the right of Q. The number of distinct seating arrangements possible is:

- A. 2
- B. 4
- C. 6
- D. 8

37.) A function, λ , is defined by

$$\lambda(p, q) = \{(p - q)^2, \text{ if } p \geq q,$$

$$p + q, \text{ if } p < q.$$

The value of the expression is: $\frac{\lambda(-(-3+2), (-2+3))}{(-(-2+1))}$

- A. -1
- B. 0
- C. 16/3
- D. 16

38.) Humans have the ability to construct worlds entirely in their minds, which don't exist in the physical world. So far as we know, no other species possesses this ability. This skill is so important that we have different words to refer to its different flavors, such as imagination, invention and innovation.

Based on the above passage, which one of the following is TRUE?

- A. No species possess the ability to construct worlds in their minds
- B. The terms imagination, invention and innovation refer to unrelated skills
- C. We do not know of any species other than humans who possess the ability to construct mental worlds
- D. Imagination, invention and innovation are unrelated to the ability to construct mental worlds

39.) The individuals _____ participated in the protest represented various societal groups.

- A. whose
- B. which
- C. who
- D. whom

40.) For a regular polygon having 10 sides, the interior angle between the sides of the polygon, in degrees, is:

- A. 396
- B. 324
- C. 216
- D. 144

41.) Which one of the following numbers is exactly divisible by $(11^{13}+1)$?

- A. $11^{26}+1$
- B. $11^{33}+1$
- C. $11^{39}-1$
- D. $11^{52}-1$

42.) oasis is to sand as island is to _____

Which one of the following options maintains a similar logical relation in the above sentence?

- A. Stone
- B. Land
- C. Water
- D. Mountain

43.) The importance of sleep is often overlooked by students when they are preparing for exams. Research has consistently shown that sleep deprivation greatly reduces the ability to recall the material learnt. Hence, cutting down on sleep to study longer hours can be counterproductive.

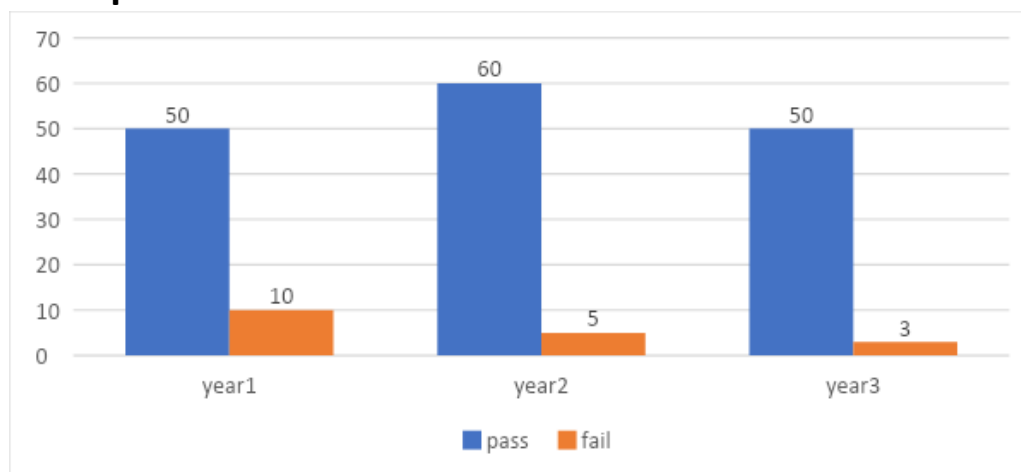
Which one of the following statements is the CORRECT inference from the above passage?

- A. Sleeping well alone is enough to prepare for an exam. Studying has lesser benefit.
- B. Students are efficient and are not wrong in thinking that sleep is a waste of time.
- C. If a student is extremely well prepared for an exam, he needs little or no sleep.
- D. To do well in an exam, adequate sleep must be part of the preparation.

44.) Seven cars P, Q, R, S, T, U and V are parked in a row not necessarily in that order. The cars T and U should be parked next to each other. The cars S and V also should be parked next to each other, whereas P and Q cannot be parked next to each other. Q and S must be parked next to each other. R is parked to the immediate right of V. T is parked to the left of U. Based on the above statements, the only INCORRECT option given below is:

- A. There are two cars parked in between Q and V.
- B. Q and R are not parked together.
- C. V is the only car parked in between S and R.
- D. Car P is parked at the extreme end.

Q 45>The number of students passing or failing in an exam for a particular subject is presented in the bar chart above. Students who pass the exam cannot appear for the exam again. Students who fail the exam in the first attempt must appear for the exam in the following year. Students always pass the exam in their second attempt.



The number of students who took the exam for the first time in the year 2 and the year 3 respectively, are _____.

- A. 65 and 53
- B. 60 and 50
- C. 55 and 53
- D. 55 and 48

46.) *Nostalgia* is to *anticipation* as _____ is to _____

Which one of the following options maintains a similar logical relation in the above sentence?

- A. Present, past
- B. Future, past
- C. Past, future
- D. Future, present

47.) Consider the following sentences:

- i. I woke up from sleep.
- ii. I woked up from sleep.
- iii. I was woken up from sleep.
- iv. I was wokened up from sleep.

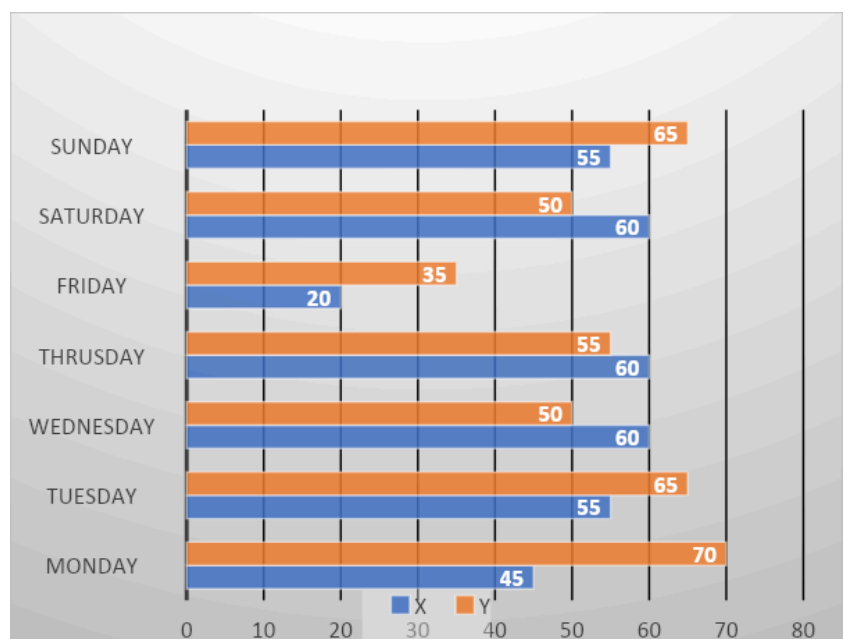
Which of the above sentences are grammatically CORRECT?

- A. (i) and (ii)
- B. (i) and (iii)
- C. (ii) and (iii)
- D. (i) and (iv)

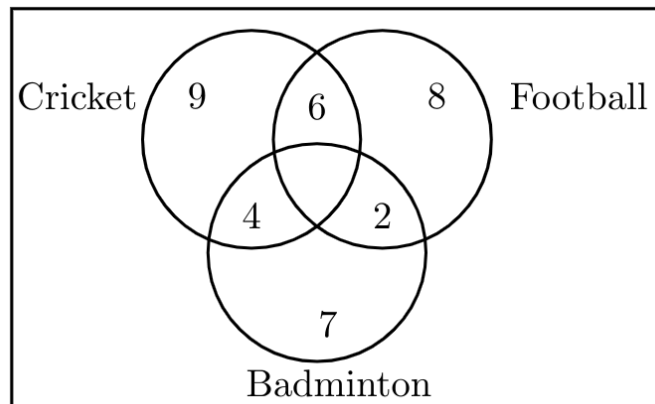
48.) The number of minutes spent by two students, X and Y, exercising every day in a given week are shown in the bar chart

The number of days in the given week in which one of the students spent a minimum of 10% more than the other student, on a given day, is

- A. 4
- B. 5
- C. 6
- D. 7



49.) In a class there are 40 students who play at least one game out of Football, Cricket and Badminton.



What percentage of students play all the three games?

- A. 4%**
- B. 5%**
- C. 8%**
- D. 10%**

50.) Ramesh's father is a pediatrician. Ram's father is a trader. Krishan's father is a school teacher. Krishan falls ill. Where should his father take him?

- A. to home**
- B. to school**
- C. to Ramesh's father**
- D. to Ram's father**

51.) On considering two relations $X(x_1, x_2, x_3)$ and $Y(x_1, y_2, y_3)$, where x_1 is the primary key of X and y_2 is the primary key of Y . If relation X consists of 110 tuples and relation Y consists of 500 tuples then the maximum number of tuples present in the operation $X \text{ EQUIJOIN } Y$ will be _____

- A. 500
- B. 400
- C. 600
- D. 300

52.) Given 4 matrices M_1, M_2, M_3, M_4 , which are having the orders of the following form $2 \times 3, 3 \times 4, 4 \times 7, 7 \times 2$, how many scalar multiplication operations are required in the optimal parenthesization_____.

- A. 90
- B. 91
- C. 92
- D. 93

53.) The time required to delete the k th element from the end of a doubly linked list given that k is a constant for a list of length n , also the length of the list is not known, and the list has reference to both first and last element of the list.

- A $O(n)$
- B $O(1)$
- C $O(\log n)$
- D Cannot be determined

54.) Which of this best describes the purpose of the port numbers that we can find in UDP or TCP headers?

- A The port number identifies a unique transport layer protocol
- B The port number identifies the version of the transport layer protocol
- C The port number is decremented at every hop
- D The port number helps the receiving kernel identify a process/socket for the packet

55.) Suppose E and F are two events of a random experiment. If the probability of occurrence of E is $\frac{1}{5}$ and the probability of occurrence of F given E is $\frac{1}{10}$, then probability of non-occurrence of at least one of the events E and F is

- A $\frac{1}{8}$
- B $\frac{1}{2}$
- C $\frac{49}{50}$
- D $\frac{1}{50}$

56.) The number of comparisons among the key elements in Bubble sort in the best case is given by

- A $\Theta(n \log n)$
- B $\Theta(n)$
- C $\Theta(n^2)$
- D None of the above

57.) Station A needs to send a message consisting of 12 packets to station B using a sliding window (window size 4) and go back n error control strategy. All packets are ready and immediately available for transmission.

If every 6th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the message to B?

A 25

B 20

C 23

D 26

58.) If an element in a binary heap is stored in the position i and the root is at position 1, then where is the parent stored?

A $\text{Floor}(i/2)$

B $\text{Ciel}(i/2)$

C $1 + \text{floor}(i/2)$

D $2i$

59.)

```
#include <stdio.h>
void foo(int[][3]);
int main(void)
{
    int i;
    int a[3][3] = { {1, 2, 3}, {4, 5, 6}, {7, 8, 9} };
    foo(a);

    for(i=0 ; i<=2 ; i++)
    {
        printf("%d ", a[2][i]);
    }
    return 0;
}
```

```
void foo(int b[][3])
{
    ++b;
    b[1][1] = 9;
}
```

A 4 5 6

B 7 8 9

C 4 9 6

D 7 9 9

60.)An organization is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets. Find the last allocatable addresses in subnet 32 ?

A 211.17.180.255

B 211.17.180.254

C 211.17.180.249

D 211.17.180.1

61.) How many disk blocks are required to keep a list of free disk blocks in a 16 GB hard disk with 1 kB block size using a linked list of free disk blocks?

Assume that the disk block number is stored in 32 bits.

- A 1024 blocks.
- B 16794 blocks
- C 20000 blocks
- D None of the above

62.) $\lim_{x \rightarrow 1} \frac{\log x}{x-1} = \underline{\hspace{2cm}}$

- A. 0
- B. 1
- C. -1
- D. 2

63.) Consider a relation R(P, Q, R, S,T) with the Functional dependency set F given as:

P → Q

P → S

Q → R

QR → S

QT → p

Which of the following statements is true about the above given relation and FDs?

- A The FD QR → S violates the BCNF, but it is in 3NF
- B The FD P → Q violates both BCNF and 3NF
- C Functional dependency QT → P is in BCNF
- D Q → R is not in BCNF but it is in 3NF

Q64.) Consider the following statements S1 and S2 :

S1 : If an undirected graph with n nodes has fewer than n edges, it cannot have a cycle

S2 : A simple graph without cycles is always a tree

Which of the following are correct ?

A S1 is True, S2 is False

B S1 is False, S2 is True

C Both S1 and S2 are False

D Both S1 and S2 are True

Q65.) Which of the following statements are True?

I. Foreign key in a relation is to ensure the NULL and Domain integrity.

II. The referenced and the referencing table have one-to-many relationship in between them.

A Only I

B Only II

C Both I and II

D Neither I nor II

66.) Arrange the following functions in increasing order of time complexity.

$f_1(n)=n!$

$f_2(n)=n^2\log(n)$

$f_3(n)=n^3\log(\log(n))$

$f_4(n)=n^2$

A f_4, f_3, f_2, f_1

B f_4, f_3, f_1, f_2

C f_4, f_2, f_3, f_1

D f_4, f_2, f_1, f_3

Q67.) Consider the following fragment of code. If the head of a non-empty linked list is passed to the function fun as a parameter (consider the first element as an element at odd location).

```
int fun(struct node *t)
{
    int i=1,sum=0;
    while(t)
    {
        if(i%2==0)
        {
            sum=sum+t->data;
        }
        else
        {
            printf("%d \t",t->data);
        }
        t=t->next;
        i++;
    }
    return sum;
}
```

Which of the following statements is TRUE

A The function returns the sum of all elements of the linked list .

B The function returns the sum of elements of the linked list which are at odd locations.

C The function prints the elements of the linked list which are at even locations.

D The function prints the elements of the linked list which are at odd locations.

68.) A TCP connection has just been established between two computers (A and B). Assume: Round trip time (RTT) = 100 ms, Congestion window threshold = 4 [segments], Receiver's advertised window = 12 [segments]. What is the least amount of time in seconds before the sender reaches full window size ?

- A. 1
- B. 2
- C. 3
- D. 4

69.) #include<stdio.h>
int main()
{
int j,c=1,d=1,e=1;
for(int i=1; i<=256; i=i*2)
{
j=1,d=1,e++;
while(j<=256)
{
j=j*3 ;
c++,
d++;
}
}
printf("%d %d %d",c,d,e);
return 0;
}
A 56 8 11
B 55 7 10
C 72 8 9
D None of the above

70.) How many letters this code will print?

```
#include <stdio.h>
```

```
#include<string.h>
```

```
int main()
```

```
{
```

```
    int count=0;
```

```
    int i, n;
```

```
    char *x = "modest";
```

```
    n = strlen(x);
```

```
    for(i=0; i<n; ++i)
```

```
    {
```

```
        printf("%s\n",x);
```

```
        x++;
```

```
        count++;
```

```
    }
```

```
    printf("%d",count);
```

```
    return 0;
```

```
}
```

A. 21

B. 20

C. 19

D. 23

71.) Which of the following statements is false?

A A maxterm is the sum of the literals within the boolean expression.

B For any two input binary variables A and B, if AB represents one of the minterm, then $A' + B'$ would be its corresponding maxterm.

C For any two input binary variables A and B, if $A+B$ represents one of the maxterm, then $A' + B'$ would be its corresponding minterm.

D None of the above

72.) Consider the three file allocation strategies:

a. Indexed files allocation

b. Linked files allocation

c. Contiguous allocation

You have a file system where the most important criteria is the performance of sequential access to very large files.

Rank the three strategies in order of best to worst.

A a, b, c

B b, a, c

C c, a, b

D None of the above

73.)The time complexity of the travelling salesman problem using dynamic programming is

A $O(n^2 2^n)$

B $O(n 2^n)$

C $O(n!)$

D None of the above

74.) The odds against a certain event are 5 to 2, and the odds in favor of another event independent of the former are 6 to 5. Find the chance that one at least of the events will happen__

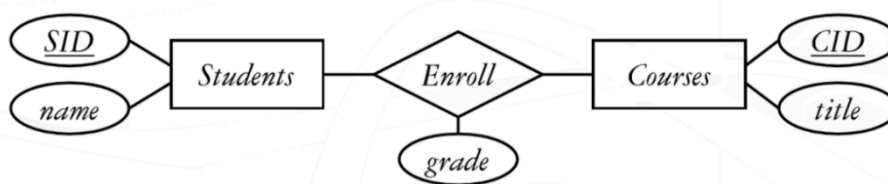
A 25/77

B 15/77

C 52/77

D None of the above

75 Given the E/R diagram below



Which of the following relations does not appear in the relational model representation of the diagram?

A. Courses (CID, title)

B. Students (SID, name)

C. Students (SID, name, CID)

D. Enroll (SID, CID, grade)

76.) The Boolean expression $=A'B'C'D + A'BCD' + AB'C'D + ABC'D'$ can be minimized to

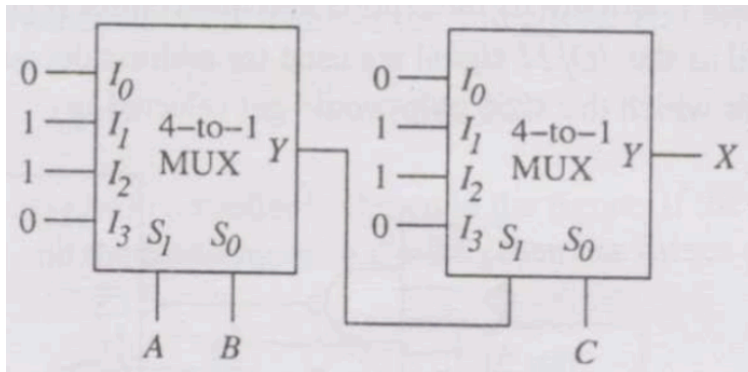
A. $Y = A'B'C'D + A'BC' + AC'D$

B. $Y = A'B'C'D + BCD' + AB'C'D$

C. $Y = A'B'CD' + B'C'D + AB'C'D$

D. $Y = A'B'CD' + B'C'D + ABC'D'$

77.) In the following circuit, XX is given by



- A. $X = AB'C' + A'BC' + A'B'C + ABC$
- B. $X = A'B'C + AB'C + ABC' + A'B'C'$
- C. $X = AB + BC + AC$
- D. $X = A'B' + B'C' + A'C'$

78.) Suppose that X is a random variable where:

$P(X=-3)=1/3$ $P(X=0)=1/3$ $P(X=3)=1/3$. and let $Y = X^2$.

What will be the value of $E[Y]$?

- A. 6
- B. 0
- C. $2/3$
- D. 1

79.) To delete a node X with 2 non-null children in a BST, we replace the node X with the minimum node Y from X's right subtree. Which of the following is true about the node Y?

- A. Y is always a leaf node
- B. Y is always either a leaf node or a node with empty left child
- C. Y may be an ancestor of the node
- D. Y is always either a leaf node or a node with empty right child

80.) Let $\langle P_1, P_2, \dots, P_n \rangle$ be the ONLY safe state of a system. Then which of the following is ALWAYS true ?

- A. If P_2 is the first process to request for some resources then P_2 always have to wait.
- B. P_1 's requests will always be satisfied without even checking if resulting state (after processing P_1 's requests) is in safe state or not.
- C. Processes requests will always be satisfied in the same order as safe state sequence and we can not satisfy any out of order process request.
- D. We can never satisfy any request coming from P_n without satisfying request from P_1 .

81.) Consider a machine with a physical memory of 8 GB, a page size of 8 KB, and a page table entry size of 4 bytes. How many levels of page tables would be required to map a 46bit virtual address space if every page table fits into a single page?

- A. 3
- B. 2
- C. 1
- D. 4

82.) Suppose that three processes exist in the ready state of the system, as described table below.

Suppose that the system uses preemptive, round-robin scheduling and that T11 is running when the quantum expires.

Process	Threads within the process
P1	T11 T12
P2	T21 T22
P3	T31

If the threads are implemented entirely at the user level (with no support from the operating system), which threads might not possibly execute at the beginning of the next quantum?

- A. T12
- B. T21
- C. T22
- D. T31

83.) Consider the following different possible pseudo-code taken from two processes, where A, B are blocks of code, and S is a semaphore initialized to 0.

Which of the following sequence ensures that A always runs BEFORE B?

- | | |
|---------------|------------|
| A. Process 1: | Process 2: |
| A; | up(S); |
| down(S); | B; |
| | |
| B. Process 1: | Process 2: |
| A; | down(S); |
| up(S); | B; |
| | |
| C. Process 1: | Process 2: |
| down(S); | down(S); |
| A; | B; |
| up(S); | up(S); |
| | |
| D. Process 1: | Process 2: |
| up(S); | down(S); |
| A; | B; |

84.) Choose all statements which are guaranteed to be false
if $f = n(\sin(n) + 2)$

- A. $f = \Omega(1)$
- B. $f = \Theta(n)$
- C. $f = O(n^2)$
- D. None of the other options are false

85.) Suppose the 4000 -byte IP packet run into a link of MTU =1500 bytes first and then a link of MTU =900 bytes next. What is FALSE about fragments sent out on the MTU =900 bytes link?

- A. Offset of last fragment will be 3840
- B. All Fragments has same ID number
- C. There will be total 6 fragments
- D. Total length of first, third and fifth fragments are same

86.) A message is a sequence of two types of signals: of type A which last for 1 second and of type B which last for 2 seconds. E.g., the message ABAAB lasts 7 seconds. How many different messages last 10 seconds?

- A. 89
- B. 90
- C. 91
- D. 92

87.) A fortune teller A's prediction is correct in 75% of the cases and that for B is correct in 80% of the cases then the probability that prediction by both of them about a person do not match is

A $7/20$

B $3/20$

C $2/7$

D $5/7$

88.) Create Table A(A1 int NOT NULL Primary key,A2 INT)

Create Table B(B1 int NOT NULL primary key,

B2 int references A

on delete cascade on update no action);

Create Table C(C1 int NOT NULL Primary key,

C2 int references B

on delete cascade on update no action);

Insert into A values (1, 1), (2, 1), (3, 1), (4, 2), (5, 2), (6, 3);

Insert into B values (1, 1), (2, 1), (3, 1), (4, 2), (5, 2), (6, 4);

Insert into C values (1, 1), (2, 1), (3, 2), (4, 3), (5, 4), (6, 6);

Assume the table A, B, and C were created and populated with data as above. Choose the correct answer that represents what is in relation C after performing “ delete from A; ”?

A (1, 1), (2, 1), (3, 2), (4, 3), (5, 4), (6, 6)

B (3, 2), (4, 3), (5, 4), (6, 6)

C (5, 4), (6, 6)

D No tuples

89.) The function fork is called once, but returns _____

A Once

B Twice

C Thrice

D None of the above

90.) Determine the time complexity of the following source code.

```
#include <stdio.h>
int main(void) {
    int n = 1000000000;
    for(int i=n; i>=10; i--)
    {
        int j=1 ;
        while(i<(n/10))
        {
            printf("Hello world");
            j/= 10;
        }
    }
    return 0;
```

- A. N
- B. N^2
- C. N^3
- D. The function will not terminate

91.) Let $A = \{a, b, c\}$

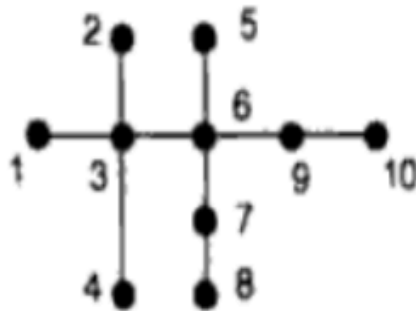
And let $R = \{ (a,a), (b,b), (c,c), (a,b), (b,c), (b,a), (c,b) \}$ be a relation on A

Which of the following is correct?

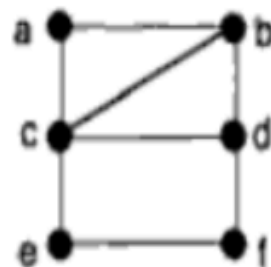
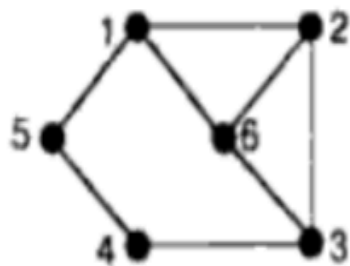
- A R is not reflexive relation
- B R is symmetric relation
- C R is transitive relation
- D R is equivalence relation

Q.)92 FOR EACH OF THE FOLLOWING, DETERMINE WHETHER THE GRAPH ISN THE GIVEN PAIR IS ISOMORPHIC?

I.



II.



A Pair I is isomorphic, Pair II is not

B Pair I is not isomorphic, Pair II is isomorphic

C Both Pair I and Pair II is isomorphic

D Neither Pair I, nor Pair II is isomorphic

93.) In a simple, connected graph on 6 vertices, the degrees of 5 vertices are 1, 2, 3, 4, 5 respectively. The degree of the 6th vertex is

- A 3**
- B 2**
- C 1**
- D 4**

94.) Consider the address 10.188.31.0/23, the possible maximum number of subnets with as many as 30 hosts each. Which is the first block address.

- A 10.188.31.0/27**
- B 10.188.31.0/26**
- C 10.188.31.0/28**
- D 10.188.31.0/29**

95.) A UNIX filesystem has 2-KB blocks and 4-byte disk addresses. Each i-node contains 10 direct entries, one singly-indirect entry and one doubly-indirect entry

What is the maximum file size approximately?

- A 409 MB**
- B 582 MB**
- C 486 MB**
- D 513 MB**

```
96.) #include<stdio.h>

int rec(int n)
{
    if(n>=0 && n<=2)
    {
        return n;
    }
    return rec(n-1)+2*rec(n-2)+rec(n-3);
}

int main()
{
    printf("%d", rec(5));
    return 0;
}
```

A. 18

B. 19

C. 20

D. 21

97.)

```
#include <stdio.h>
```

```
void foo(int);
```

```
int main()
```

```
{
```

```
    foo(0);
```

```
    return 0;
```

```
}
```

```
void foo(int inp)
```

```
{
```

```
    if(inp==5) return;
```

```
    ++inp;
```

```
    foo(inp);
```

```
    printf("%d ",inp);
```

```
}
```

Output of the following code?

A 5 5 5 5 5

B 5 4 3 2 1

C 1 2 3 4 5

D None of the above.

98.) The maximum number of nodes possible in B+ tree with height 5 and order 3 is _____(NOTE: height of the first node here is 0)

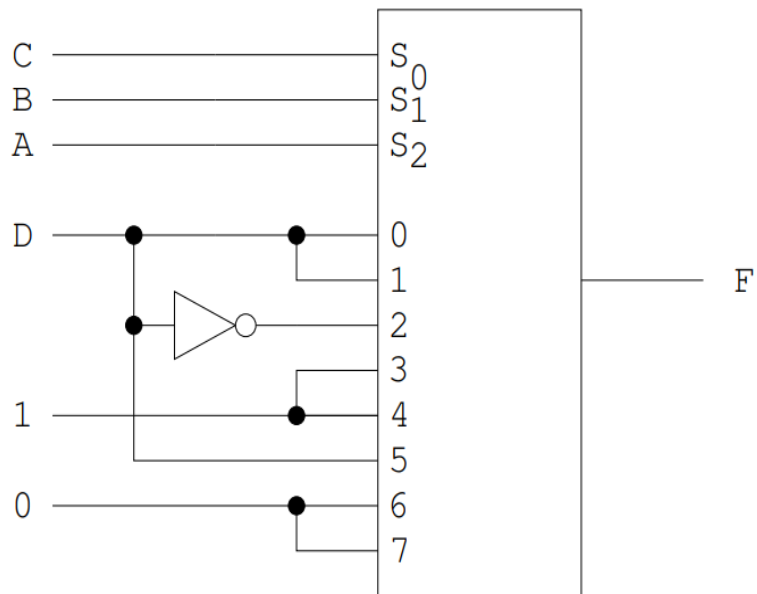
A. 364

B. 365

C. 366

D. 367

99.)



The figure above implements the Boolean function:

- A. $F(A,B,C,D)=\Sigma(0,2,3,5,6,8,12,13)$
- B. $F(A,B,C,D)=\Sigma(1,3,4,6,7,8,12,13)$
- C. $F(A,B,C,D)=\Sigma(0,2,4,5,7,8,9,11)$
- D. $F(A,B,C,D)=\Sigma(1,3,4,6,7,8,9,11)$

100.) Consider these 2 statements -

- S1: If A is invertible, then A and A^{-1} have the same eigenvectors.
- S2: If A is invertible, then A and A^{-1} have the same eigenvalues.

Which of the following is the correct option?

- A. S1 is correct but S2 is incorrect
- B. S1 is incorrect but S2 is correct
- C. Both are correct
- D. Both are incorrect

1.) SOLUTION: D

The sentence is discussing something being overpriced or too costly to be considered reasonable or just.

The correct choice is:

1.) The fare is excessively elevated to be deemed fair.

Explanation:

- Fare: Refers to the price one pays for transportation (like a bus or train ticket).
- Fair: Refers to something being just, reasonable, or acceptable in terms of value or quality.

In this context, the sentence suggests that the transportation cost (fare) is too high to be considered reasonable or just (fair).

2.) SOLUTION C

Given that,

$$x^2+2x+6=0$$

now as r is root of the equation , so we can write

$$r^2+2r+6=0$$

$$\text{or , } r^2=-2r-6$$

we have to find

$$(r+2)(r+3)(r+4)(r+5) \quad [\text{ where } r \text{ is the root of the equation }]$$

$$= (r^2+5r+6)(r^2+9r+20)$$

$$= (-2r-6+5r+6)(-2r-6+9r+20)$$

$$= 3r(7r+14)$$

$$= 21(r^2+2r) \quad [(r^2+2r)=-6]$$

$$= 21*(-6)$$

$$= -126$$

answer is (-126).

3.) SOLUTION B

Given: $(100008)^{12499}$

Now,

$$8^1=8$$

$$8^2=64$$

$$8^3=512$$

$$8^4=4096$$

$$8^5=32768....$$

So, observe the value at unit's place, it is repeating after 4 numbers.

So, now

$$12499 \bmod 4 = 3$$

Hence, unit place will have 2.

4.) SOLUTION D

Upon the final fold, if the two equal sides measure 10 cm each, then the length of the base of the triangle before the last fold (i.e., the base of the last triangle shown in the figure) should be twice that length, i.e., 20 cm. Hence, the base of the square piece of paper is 20 cm. Therefore, the area is $20^2=400$.

5.) SOLUTION B

1)HARDIKPANDYA

ROHITSHARMA

HARA

2) HARDIKPANDYA

ROHITSHARMA

HIAA

2) HARDIKPANDYA

ROHITSHARMA

RIAA

There are 3 possible longest subsequence with length 4.

Hence. $4+3*5$

19

6.) SOLUTION C

Let's supposed added number be X.

Then, the equation will be

$$(2+x)/(5+x)=5/6$$

Required number =x=13.

Hence, option C is correct.

7.) SOLUTION A

For the given series pattern are as follows:

- **$10+9=19$**
- **$19+11=30$**
- **$30+13=43$**
- **$43+15=58$**
- **$58+17=75$**

8.) SOLUTION D

All except Veil cover the head, while veil covers the face.

9.) SOLUTION A

To find the number of trailing zeros in 25! (25 factorial), we need to determine the number of times 10 can be factored out from the product of all positive integers up to 25.

A trailing zero is produced by a combination of 2 and 5. Since 2s occur more frequently than 5s in the sequence of natural numbers, we only need to count the number of 5s.

To count the number of 5s in 25!:

$$25/5 = 5$$

$$25/25 = 1$$

Adding these counts, we get: $5 + 1 = 6$

So, 25! has 6 trailing zeros.

10.) solution C

$$1 \text{ million} = 1000000$$

so no of digits should be in the range of 1 to 6

for filling one digit we have 3 options (either 0,7,8)

so total possibility for 6 digits $= 3^6 = 729$

out of which its count 000000 which is equal to 0, can't be considered for this question.

so, total numbers $= 729 - 1 = 728$

11.) Solution B

**1.) group with one Indian – 2C_1 (choose 1 Indian out of three) * 2^4
(for Chinese people, each Chinese have 2 possibility either he
present in group or not) = $2 * 2^4 = 32$**

2.) group with 2 Indian = ${}^2C_2 * 2^4 = 16$

total group with at least one Indian = $32+16= 48$

OR

Total Group – group without Indian

$$(2^6 - 1) - (2^4 - 1) = 48$$

12.) Solution D

$$(7+13) \times 3 = 20 \times 3 = 60$$

$$(12+13) \times 4 = 25 \times 4 = 100$$

$$(30+31) \times 3 = 61 \times 3 = 183$$

So, the correct answer is (D).

13.) Solution B:

**For max area, the rectangle will be a square. So, if x is the
side of the square, we have**

$$x^2 + x^2 = (2(2))^2$$

**Here, the diagonal of the square will be equal to the
diameter of the circle.**

**Solving this, we get $x = 2(2)^{1/2}$ or the area of the square as 8 sq.
Units.**

14.) SOLUTION D

Phlegmatic → CALM

15.) SOLUTION B

As opposed to “as well as” which considers only the first subject, “and” considers both the subjects before and after it and so the verb usage must be “plural”. So, (i) is correct and (ii) is wrong.

Rohit’s “family” is correct as one can have only one family. So, (iv) is correct.

16.) SOLUTION D

Given that, $p \vee q = p \vee q$, and $p \wedge q = p \wedge q$

Then, $(9 \vee (6 \wedge 7)) \vee (7 \wedge (6 \vee 4)) = (9 \vee 42) \vee (7 \wedge 2) = -33 \vee 14 = -33 - 14 = -47$.

So, the correct answer is D

17.) SOLUTION C

It is given that, four persons W, X, Y and Z are to be seated in a row. Y should not be seated at the second position from the left end of the row.

Now, we can make the possible arrangements.

- From the left end, the second position can be filled in 3 ways.
- From the left end, the first position can be filled in 3 ways.
- From the right end, the second position can be filled in 2 ways.
- From the right end, the first position can be filled in 1 way.

∴ The number of distinct seating arrangements possible $= 3 \times 3 \times 2 \times 1 = 18$.

18.) SOLUTION A

This means that the author considers rehearsal either equally or more important for public speakers than for musicians and actors. So, options B and C are clearly false. Option D is not true as the author is not comparing musicians and actors.

19.) SOLUTION B

Given that, five persons P,Q,R,S and T are to be seated in a row.

- P and T cannot be seated at either end of the row.
- P should not be seated adjacent to S.
- R is to be seated at the second position from the left end of the row.

Now, the five persons can be seated in the below arrangements:

- First we fix the position of R, and now we have two choices for P and T, then place the S and Q in the remaining positions.
 - SRPTQ
 - QRPTS
 - SRTPQ

The number of distinct seating arrangements possible is =3.

20.) Solution A

“The number of candidates” is singular and refers to the actual number. So, “is staggering” is correct.

“A number of candidates” is plural as here the number refers to a group of possibilities (from $\{1,2,3,4,\dots N\}$) and is hence plural. So, “are appearing” is correct.

21.) SOLUTION B.

Given that, a digital watch X beeps every 40 seconds while watch Y beeps every 42 seconds.

Now, digital watch X and Y beep together every $=\text{LCM}(40,42)=840$ seconds $=14$ minutes.

First, they beeped together at 10AM. Then the immediate next time that they will beep together is $=10\text{AM}+14$ minutes $=10:14\text{AM}$.

So, the correct answer is B

22.) Solution D

The front door of Ms. Sudo's house faces East. Then the back door of Ms Sudo house faces West (the back door that is situated directly opposite to the front door).

$\therefore$ The direction of the point Ms Sudo is now located at with respect to the starting point is North-West.

So, the correct answer is (D).

23.) SOLUTION C

C draw the Venn diagram and check

24.) SOLUTION B

It is given that from a box containing 15 blue balls and 45 black balls, 2 balls are selected randomly without replacement.

Probability that, the first selected is a blue ball and the second selected is a black ball

$$=(15C1)(45C1)/(60*59)$$

$$=(15*45)/(60*59)=45/236.$$

25.) Solution D

The air travel industry is facing a crisis, as the resulting quarantine requirement

The usage of “resulting” shows that the second statement is a result/effect of the first one.

26.) SOLUTION D

Hardly/barely is an adverb and means “almost not true”.

Hardly/barely comes before the main verb of a sentence, but when there is a modal or auxiliary verb, hardly usually comes after it:

- After his surgery, Raja could hardly walk.
 - Here, could is the auxiliary verb and hardly/barely should follow it.

27.) SOLUTION D

Let the radius of the circle be r cm.

The area of triangle $= \frac{1}{2} \times \text{base} \times \text{height}$

The area of the right triangle $MON = \frac{1}{2} \times r \times r = 50$

$$\Rightarrow r^2 = 100$$

$$\Rightarrow r = 10 \text{ cm}$$

Now, the area of circle $= \pi r^2 = \pi (10)^2 = 100 \pi \text{ cm}^2$.

28.) SOLUTION B

The value of the expression $\Delta 3 \oplus 3 \Delta ((4 \otimes 2) \nabla 4)) = +3 - 3 + ((4 \div 2) \times 4) = 0 + 8 = 8$.

29.) SOLUTION C

- A. The people in the state did not consume leafy vegetables earlier
No. "The increased consumption of leafy vegetables in the recent months". Talk is about increased consumption and this cannot imply "no consumption" earlier.
- B. Consumption of leafy vegetables may not be the only indicator of healthy lifestyle
May be true but nothing conclusive in the given passage.
- C. Leading a healthy lifestyle is related to a diet with leafy vegetables
True. "The increased consumption of leafy vegetables in the recent months is a clear indication that the people in the state have begun to lead a healthy lifestyle"
- D. The people in the state have increased awareness of healthy hazards causing by consumption of junk foods
May be true. But nothing conclusive in the given passage.

30.) Solution : C.

Oxpeckers and rhinos exhibit a symbiotic bond in their natural habitat. While oxpeckers alert rhinos to the presence of potential threats like poachers, thereby potentially safeguarding the rhinos, they also consume ticks feeding on rhinos, providing oxpeckers with a food source.

31.) SOLUTION C

First, we can calculate the number of hens, ducks and goats in farm B.

- Number of hens in farm B = $532 \times 416 = 65$
- Number of ducks in farm B = $1432 \times 416 = 182$
- Number of goats in farm B = $1332 \times 416 = 169$

Initially, the number of hens, ducks and goats in farm A are 65, 91 and 169 respectively.

All the hens, ducks, and goats are sent from farm B to farm A.

Therefore, in farm A,

- Number of hens = $65 + 65 = 130$
- Number of ducks = $91 + 182 = 273$
- Number of goats = $169 + 169 = 338$

∴ The required ratio = $130:273:338 = 10:21:26$

32.) SOLUTION C

Given that the total number of employees at the rank of executives across all companies is 10,000.

- The number of employees in the company C2 = $5/100 \times 10000 = 500$
- The number of employees in company C2, who has management degree = $1/5 \times 500 = 100$
- The number of employees in the company C5 = $20/100 \times 10000 = 2000$
- The number of employees in company C2, who has management degree = $9/10 \times 2000 = 1800$

∴ The total number of management degree holders among the executives in companies C2 and C5 together = $100 + 1800 = 1900$.

So, the correct answer is (C).

33.) Solution D

Given that, five persons P, Q, R, S, and T are sitting in a row.

1. Q and R are separated by one person.
2. S should not be seated adjacent to Q.

We can follow the above two conditions we get the following sitting orders:

- First fix the position of Q, and R, we get,
 - QPRST
 - QPRTS
 - QTRSP
 - QTRPS
- Again, fix the position of Q, and R, we get,
 - PQTRS
 - TQPRS
- Again fix the position of Q, and R, we get,
 - SPQTR
 - STQPR

Now, we can fix the positions interchangeably, and get the following sitting orders:

- First fix the position of R, and Q, we get,
 - RPQTS
 - RTQPS
- Again fix the position of R, and Q, we get,
 - SRPQT
 - SRTQP
- Again fix the position of R, and Q, we get,
 - SPRTQ
 - STRPQ
 - PSRTQ
 - TSRPQ

∴ The number of distinct seating arrangements possible =16.

34.) SOLUTION D

Since “than” is followed we need the comparative form of “easy” which is easier.

35.) SOLUTION B

Given that, $p \oplus q = p^2 + q^2 / pq$ and $p \odot q = p^2 / q$

Performing these operations we get,

$$x \oplus y = 2 \odot 2$$

$$\Rightarrow x^2 + y^2 / xy = 2^2 / 2$$

$$\Rightarrow x^2 + y^2 / xy = 2$$

$$\Rightarrow x^2 + y^2 = 2xy$$

$$\Rightarrow x^2 + y^2 - 2xy = 0$$

$$\Rightarrow (x - y)^2 = 0$$

$$\Rightarrow x = y$$

36.) SOLUTION C

Given that, four persons P, Q, R and S are to be seated in a row.

- P and R cannot sit adjacent to each other.
- S should be seated to the right of Q.

Now, the four persons can be seated in the below arrangements:

- QPSR
- QRSP
- PQSR
- RQSP
- PQRS
- RQPS

∴ The number of distinct seating arrangements possible = 6.

37.) SOLUTION B

The value of the expression

$$\frac{\lambda(-(-3+2),(-2+3))}{(-(-2+1))} = \frac{\lambda(-(-1),1)}{-(-1)} = \frac{\lambda(1,1)}{1} = \frac{0}{1} = 0$$

So, the correct answer is B

38.) SOLUTION C

A. No species possess the ability to construct worlds in their minds

False. "Humans have the ability to construct worlds entirely in their minds" and humans is a species.

B. The terms imagination, invention and innovation refer to unrelated skills

False. "This skill is so important that we have different words to refer to its different flavors, such as imagination, invention and innovation" meaning the three terms are related to the same skill.

C. We do not know of any species other than humans who possess the ability to construct mental worlds

True. "So far as we know, no other species possesses this ability."

D. Imagination, invention and innovation are unrelated to the ability to construct mental worlds

False. "This skill is so important that we have different words to refer to its different flavors, such as imagination, invention and innovation."

39.) SOLUTION C

To refer to "people" which is plural we must use "who" and not "whom".

"whose" is used to refer to some object or property being associated with a person which is not applicable here.

"which" and "that" are the relative pronouns for animals and objects (not humans).

Correct option: C

40.) SOLUTION D

The interior angle between the sides of the polygon = $\frac{(n-2) \times 180}{n}$

where n= number of sides of the polygon.

Here, n=10, therefore the interior angle between the sides of the polygon = $\frac{(10-2) \times 180^\circ}{10}$

= $8 \times 18^\circ = 144^\circ$.

So, the correct answer is D

41.) SOLUTION D

We know that,

- $x^n - y^n$ is divisible by $x+y$, if n is even.
- $x^n - y^n$ is divisible by $x-y$, if n is odd.
- $x^n + y^n$ is divisible by $x+y$, if n is odd.

Now, we can check each option.

A. $11^{26}+1=(11^{13})^2+1^2$ is divisible by $11^{13}+1$, if 2 is odd.

B. $11^{33}+1=(11^{11})^3+1^3$, here $11^{11}+1 \neq 11^{13}+1$.

C. $11^{39}-1=(11^{13})^3-1^3$, here $11^{13}-1 \neq 11^{13}+1$.

D. $11^{52}-1=(11^{13})^4-1^4$ is divisible by $11^{13}+1$, if 4 is even.

So, the correct answer is (D).

42.) SOLUTION C

Oasis: a fertile spot in a desert, where water is found

So, we find oasis surrounded by sand.

Similarly, island is surrounded by water.

Correct option: C.

43.) SOLUTION D

A. Sleeping well alone is enough to prepare for an exam.

Studying has lesser benefit.

No where in the passage this is implied. The given passage only says that "sleeping well" is a necessity. It doesn't say sleeping well is "sufficient" to prepare for an exam.

B. Students are efficient and are not wrong in thinking that sleep is a waste of time.

This is contradictory to the given passage.

C. If a student is extremely well prepared for an exam, he needs little or no sleep.

False. "sleep deprivation greatly reduces the ability to recall the material learnt."

D. To do well in an exam, adequate sleep must be part of the preparation.

True. "cutting down on sleep to study longer hours can be counterproductive"

Correct option D.

44.) SOLUTION A

- 1. The cars T and U should be parked next to each other.
This means we can park these two cars in two ways, either TU or UT.**
- 2. The cars S and V also should be parked next to each other.
This means we can park these cars also in two ways, either SV or VS.**
- 3. The cars P and Q cannot be parked next to each other.
This means we can not park the cars as PQ or QP.**
- 4. The cars Q and S must be parked next to each other.
This means we can park these two cars in two ways, either QS or SQ.**
- 5. R is parked to the immediate right of V.
This means we can park these two cars in only one way VR.**
- 6. T is parked to the left of U.
This is not “immediately left” but combining with statement 1, “immediately left” follows. So, we can park these two cars in only one way TU.**

Now we can combine the statements (2)&(4) and get two possible ways for cars Q,S and V.

- QSV**
- VSQ**

But VSQ is not possible because it violates statement 5.

Now, using statement 5, we get the possible car parking sequence QSVR.

Now, we can check each and every option.

- A. There are two cars parked in between Q and V, it is incorrect because only S is in between Q and V.**

- B. Q and R are not parked together, it is correct.
- C. V is the only car parked in between S and R, it is correct.
- D. Car P is parked at the extreme end.

For this option, we can consider all possible combinations of car parking which are only four as given below. (P cannot come near Q due to statement 3)

- o QSVRPTU
- o QSVRTUP
- o TUQSVRP
- o PTUQSVR

Here, we can see that car P can be parked at the extreme (either left or right) end in three out of the four possible combinations. That is, though this is not a guarantee, it is still a possibility.

So, the correct answer is A.

45.) SOLUTION D

The total number of students in Year 2 = $60(\text{pass}) + 5(\text{fail}) = 65$

The number of students, who failed in the exam in Year 1, and appeared in Year 2 = 10

∴ The total number of students in Year 2 who appeared for the first time = $65 - 10 = 55$

Similarly, the total number of students in Year 3 = $50(\text{pass}) + 3(\text{fail}) = 53$

The number of students, who failed in the exam in Year 2, and appeared in Year 3 = 5

∴ The total number of students in Year 3 who appeared for the first time = $53 - 5 = 48$

So, the correct answer is (D).

46)solution C

C. Past, Future

As Nostalgia: recalling memories (of past) and Anticipate: Foresee (see the future)

47.)) SOLUTION B

wake, woke, woken is the correct sequence for present, past and past participle forms of the word “wake”. So, only (i) and (iii) are correct.

48 SOLUTION C

We can calculate the number of days in the given week in which one of the students spent more time than the other.

If X is more than 10% of Y then Y should be minimum and vice-versa.

- On Monday, the required percentage = $(\frac{70-45}{45}) \times 100\% = 55.55\%$
- On Tuesday, the required percentage = $(\frac{65-55}{55}) \times 100\% = 18.18\%$
- On Wednesday, the required percentage = $(\frac{60-50}{50}) \times 100\% = 20\%$
- On Thursday, the required percentage = $(\frac{60-55}{55}) \times 100\% = 9.09$
- On Friday, the required percentage = $(\frac{35-20}{20}) \times 100\% = 75\%$
- On Saturday, the required percentage = $(\frac{60-50}{50}) \times 100\% = 20\%$
- On Sunday, the required percentage = $(\frac{65-55}{55}) \times 100\% = 18.18\%$

∴ Total 6 days are there, when one of the students spent a minimum of 10% more than the other student.

So, the correct answer is C

49 SOLUTION D

The number of students, who plays all the three games =x

Now, $9+6+4+x+8+2+7=40$

$$\Rightarrow x+36=40$$

$$\Rightarrow x=4$$

$$\text{Percentage} = \frac{4}{40} \times 100\% = 10\%.$$

50.) SOLUTION C

Given that, Ramesh's father is a pediatrician (a doctor who deals with the diseases of children).

If Krishan falls ill. then his father takes him to Ramesh's father.

51 Solution: A

x1 is the common attribute in between relation X and Y and is a primary key of X, therefore, all 110 values of x1 in X are unique. Since X contains 110 x1 values and Y contains 500 values of x1 then these 110 unique values could be associated with 500 values of x1 in Y. But x1 in Y can have multiple entries of the same. Hence, the maximum number of tuples in the result set of R EQUIJOIN S will be 500.

52 Solution C

The Recurrence relation for getting the minimum no of matrix multiplication is

$$M[i,j] = 0 \text{ if } i=j$$

$$\min_{1 \leq k < j} \{M[i,j] = M[i,k] + M[k+1,j] + P_{i-1}P_kP_j\}$$

$$M[1,1]=0$$

$$M[2,2]=0$$

$$M[3,3]=0$$

$$M[4,4]=0$$

$$M[1,2]=2*3*4=24$$

$$M[2,3]=3*4*7=84$$

$$M[3,4]=4*7*2=56$$

$$M[1,3] = \text{Min}((M[1,1]+M[2,3] + (2*3*7)), \\ (M[1,2]+M[2,2] + (2*4*7))) = \text{Min}(84+24, 80) = 80$$

$$M[2,4] = \text{Min}((M[2,2]+M[3,4] + (3*4*2)), \\ (M[2,3]+M[4,4] + (3*7*2))) = \text{Min}(80, 84+42) = 80$$

$$M[1,4] = \text{Min}((M[1,1]+M[2,4] + (2*3*2)), \\ (M[1,2]+M[3,4] + (2*4*2)), (M[1,3]+M[4,4] + (2*7*2))) = \text{Min}(80+12, \\ 24+56+16, 80+28) = 92$$

53.) Solution B

Since k is a constant even though the length of list is now known if we start iterating from the last/tail pointer we can move forward to the k th element with a constant number of moves and the element can also be deleted by a constant number of pointer changes, therefore the time required is $O(1)$.

54.) Solution: (D)

Yes port number helps to determine for which process this packet belongs. It helps the kernel to determine the exact process.

55.) SOLUTION C

$$P(E)=1/5$$

$$P(F|E)=1/10$$

$$P(E \cup F) = P((E \cup F)') = 1 - P(E \cap F) = 1 - P(E) * P(F|E) = 1 - (1/5) * (1/10) = 49/50.$$

56.) Solution: (B)

In case of bubble sort the best case occurs when the array is already sorted

57) Solution c

1 2 3 4 5 6* 7 8 9 => 6th packet lost

6 7 8* 9 10 11 => 8th packet loss

8 9 10* 11 12 => 10th packet loss

10 11 12

58.) Solution: (A)

Take an example:

If array have 5 elements index starting from 1.

1	2	3	4	5
10	11	13	12	9

The root of element 9 on index 5 is =2

So the option a= floor(i/2) = 5 / 2 = 2

The only option matches the criteria.

59.) Solution: (D)

We are passing the address of 2d array a

In the foo function, we are incrementing b

b will start pointing to the second 1-d array.

b[1][1] represents the second value of third 1 d array.

8 is replaced by 9

Using for loop we are printing the third 1-d array.

7 9 9 will print finally.

60.) SOLUTION B

Allocatable address is the host address

The block is 211.17.180.0/24

For 32 subnets requires 5 bits, so remaining three are host bits.

The 32th subnet is 211.17.180.11111 000 = 211.17.180.240

The last host address = 211.17.180.1111 110 = 211.17.180.254

61.) SOLUTION D

No. of blocks in disk = $16 \times 2^{30} / 2^{10}$; ' = 16×2^{20} blocks

no. of block address one block can store = $(1024 \times 8) / 32 = 256$

each block have one address as pointer to next block

therefore, no. of block addresses in a block = 255

∴ No. of blocks needed to keep list of free blocks = $16 \times 2^{20} / 255 = 65794$ BLOCKS

62.) SOLUTION B

$(\log x/x - 1)$ is in 0/0 indeterminate form.

Applying L'Hopital's rule

$$(1/x)/x = 1$$

63.) Solution c

For the given functional dependency set F,

candidate keys are:

P+ = {P, Q, R, S} //not a key

PT+ = {P, Q, R, S, T} //it is a CK

QT+ = {P, Q, R, S, T} //it is a CK

ST+ = {S, T} //not a key RT- = {R, T} //not a key

Therefore, the possible candidate keys are {PT, QT}

=> For FD QR -> S: The LHS "QR" is not a super key/key and RHS is not a prime attribute, therefore, it is neither in 3NF nor in BCNF.

Hence, this statement is not true.

=> For FD P -> Q: The LHS "P" is not the super key/key, therefore, it is not in BCNF, but the RHS of FD "Q" is a prime attribute, therefore it is in 3NF. Hence, this statement is also not true.

=> For FD QT -> P: The LHS "QT" is a candidate key, therefore, it is in both BCNF and 3NF. Hence, this statement is correct/true.

=> For FD Q -> R: Neither the LHS "Q" is not the super key/key, nor the RHS "R" is a prime attribute, therefore, it is neither in 3NF nor in BCNF. Hence, this statement is also not true.

64.) Solution: C

S1 : False

A connected graph needs as many edges as vertices in order to have a cycle. But a graph consisting of a triangle and a separate isolated node has four nodes and three edges, and yet has a cycle.

S2 : False

A tree is a simple graph without cycles, but it must also be connected.

65.) Solution: (B)

Since the purpose of the foreign key is to identify a particular row of referenced table, it is generally required that the foreign key is equal to the candidate key in some row of the primary table, or else have no value the NULL value. This rule is called a referential integrity constraint. Thus, foreign keys are used to ensure referential integrity constraint and not NULL or domain integrity.

Since multiple rows in the referencing table may refer to the same row in the referenced table, in this case, the relationship between the two tables is called a one-to-many relationship between the referenced table and the referencing table.

66.) solution C

n^2 grows slower than $n^2 \log(n)$, and $n^2 \log(n)$ grows slower than $n^3 \log(\log(n))$ and $n!$ is the fastest function of all as it is a factorial function.

67.) Solution: D

Since the summation statement is in the if block with the condition $i \% 2 == 0$ the function returns the sum of even elements and the print statement is present in the else block it prints the odd elements.

68.) Solution: A

Max sender window size = 12.

First transmission : 1

Second transmission : 2

Third transmission : 4

Fourth transmission : 5

Fifth transmission : 6

sixth transmission : 7

Seventh transmission : 8

Eighth transmission : 9

Ninth transmission : 10

Tenth transmission : 11

Eleventh transmission : 12

Clearly, the sender window reaches its max size after 10RTT = 1s.

Before the start of 11 RTT window size is 12 that why we haven't taken eleventh transmission in account .

69 Solution: (B)

Variable e is representing how many times the outer loop will run.

which is 1 , 2 , 4 , 8 ,16, 32, 64 , 128, 256 = 9 times

So e++ will run 9 times

then, $e = e + 9 = 10$

Variable d is representing how many time inner loop will be executed.

Which is 1 , 3 , 9, 27, 81, 243= 6 times

So $d = d + 6 = 7$

Therefore, inner loop = $9 * 6 = 54$

$c = c + 54 = 55$

70.) Solution A

n will store the string length which is 6

%s will print the string

x will be pointing to next char of string in every iteration The code will print:

modest

odest

dest

est

st

t

Total letters = 6 + 5 + 4 + 3 + 2 + 1 = 21

71.) Solution: (C)

(i) It is true as sum of literals are called as maxterms and the product of maxterms is an POS expression.

We know that, in the SOP, method, the value '1' is replaced by the variable and '0' by its complement. In contrast, when it comes to POS a '0' is substituted by the variable and '1' by its complement.

If given minterm is AB (i.e. A = 1, and B = 1) then its equivalent POS will be A' + B' (for A = 1 and B = 1). Therefore, option (ii) is correct and option (iii) is incorrect.

72.) SOLUTION D

The order is:

1. c

2. b (linked)

3. a (indexed)

It is easy to see that. (c) is the best structure for sequential access to Very large files, since in (c) files are contiguously allocated and the next block to read is physically the next on the disk. No seek time to find the next block, and each block will read sequentially as the disk moves. Both (b) and (a) require some look up operation in order to know where the next block is.

However, (a) may be slightly more expensive, since for "very large files", multiple disk accesses are required to read the indirect blocks.

73.) Solution A

The time complexity of travelling salesman problem using dynamic programming is $O(n^2 2^n)$

74.) Solution C

$$P(-A)=5/7 \Rightarrow P(A)=2/7$$

$$P(-B)=5/11 \Rightarrow P(B)=6/11$$

Now let E be the event that at least one of them occurs

$$P(E)=P(A \cap B)+P(A' \cap B)+P(A \cap B')$$

$$=P(A)P(B)+P(A')P(B)+P(A)P(B') \text{ (as events A and B are independent)}$$

$$=(2/7)*(6/11) + (5/11)*(6/11) + (5/7)*(5/11)$$

$$=52/77.$$

75.) Solution c

The given relationship "enroll" is many to many, with both sides having partial participation.

So, A,B,D will be the relations in relational model.

76.) Solution D

Using K map, we can solve

77.) SOLUTION A

Y is $A \oplus B$

So, $X = (A \oplus B) \oplus C$

X is 1 iff Odd number of inputs are 1. So, the answer is Option A.

78.) SOLUTION A

Y is $A \oplus B$

So, $X = (A \oplus B) \oplus C$

X is 1 iff Odd number of inputs are 1. So, the answer is Option A.

79.) Solution B

We replace delete node in BST either by inorder successor or inorder predecessor, it is given in the question that we replace by minimum value in right subtree of the node which means inorder successor.

80.) SOLUTION B

Safe sequence means that you are having a sequence in that order you can ALWAYS satisfy MAXIMUM need of any resource if that is in order.

But if there is some other request that coming out of order then also, we can satisfy that request, we just need to check if resulting state is safe or not.

Let P2 requests for n number of resources which need to be maximum NEED. Then we will apply banker's algorithm to check safe state after assigning n resources.

- A. P2 do not ALWAYS need to wait, it depends is resulting state is safe.
- B. We can ALWAYS satisfy P1 request because we know a safe sequence exist starting with P1, so if we assign then resulting state is always safe.
- C. This is false.
- D. This is also false.

81 SOLUTION A

Since each PTE is 4 bytes and each page contains 8 KB, then a one-page page table would point to 2048 or 2^{11} pages, addressing a total of $2^{11} * 2^{13} = 2^{24}$ bytes.

- Depth 1 = 2^{24} bytes
- Depth 2 = 2^{35} bytes
- Depth 3 = 2^{46} bytes

So, in total, 3 levels of page tables are required.

82 Solution A

Since threads are at user level hence The OS is unaware of the threads within process P1.

Either P2 or P3 will get scheduled in next quantum so one of the thread of either P2 or P3 will get executed.

83.) Solution B

down(S) in Process 2 blocks the execution of B until A's execution and the semaphore S is up()-ed.

84.) SOLUTION D

Value of $\sin(n)$ will range from -1 to 1 , hence $\sin(n)+2$ is always bounded by a constant on both the upper and lower sides. i.e,

$$f=n(\sin(n)+2)=\Omega(cn)=O(cn)=\Theta(cn)$$

Hence D is the correct choices

85.) SOLUTION A

6 fragments

(length, ID, flag, offset)

=(900,X,1,0)

(620,X,1,110)

(900,X,1,185)

(620,X,1,295)

(900,X,1,370)

(160,X,0,480)

86.) SOLUTION A**#Different messages = #messages contains**

A	B
10	0
8	1
6	2
4	3
2	4
0	5

$$[(10A) + (8A, 1B) + (6A, 2B) + (4A + 3B) + (2A, 4B) + (5B)]$$

$$= (1) + (9!/(8!*1!)) + (8!/(6!*2!)) + (7!/(4!*3!)) + (6!/(2!*4!)) + 1$$

$$= 1 + 9 + 28 + 35 + 15 + 1 = 89$$

87.) Solution A

$$P(A) = 75/100$$

$$P(B) = 80/100$$

A, B are independent events.

We need to find the probability that both the predictions contradict
in other words $P(A' \cup B) + P(A \cup B')$

$$= P(A')P(B) + P(A)P(B')$$

$$= (1/4)1 + (3/4)(1/5) = 7/20.$$

88.) Solution: (D)

On performing the SQL query "Delete from A; ", all the content of A will be erased which will lead to the deletion of all the tuples in B which are referring to A.

Since all the tuple in B have value referred by B2, therefore, complete set of tuples will get deleted from B, this cascades in relation C also and leads to the deletion of all the tuples in C. Thus, the correct option will be (iv).

89.) Solution: (B)

To create a new process, in UNIX, the fork () system call is used. fork () creates a new context based on the context of the calling process. The fork () call is unusual in that it returns twice: It returns in both the process calling fork () and in the newly created process. The child process returns zero and the parent process returns a number greater than zero.

90.) SOLUTION D

As the while loop goes into an infinite loop the condition will be true always.

91.) SOLUTION B

Relation R is reflexive since (a, a), (b, b), (c, c) belongs to R

R is symmetric as $(x, y) \in R \Rightarrow (y, x) \in R$, that is, $xRy \Rightarrow yRx$ for all (x, y) belongs to R.

R is not transitive as (a,b) , (b,c) belongs to R but (a, c) *doesn't belong to R*

Since R is not transitive, R is not equivalence as well.

92.) Solution: (D)

. These graphs are not isomorphic. Vertex 3 (degree 4) is forced to map to either vertex c or h as they are the only two vertices in that graph with degree 4. However, the neighbours of vertex 3 have degrees 1,1,1 and 4, while the neighbours of both vertex c and h have degrees 1,1,1 and 2

.These graphs are not isomorphic. The first graph has degree sequence $\langle 3,3,3,3,2,2 \rangle$ and the second has degree sequence $\langle 4,3,3,2,2,2 \rangle$.

93.) Solution: A

Let us call the 5 vertices with known degree $v_1, v_2, \dots, v_5$, where $d(v_i) = i$. The degree of v_6 is unknown. Node v_5 is connected by an edge to every node, so the only neighbor of v_1 is v_5 . Node v_4 is connected to every node except v_1 . Therefore, the two neighbors of v_2 are v_5 and v_4 . For v_6 , we know that it is connected by an edge to v_5 and v_4 and not connected to v_1 and v_2 . The same is true for v_3 . Since the degree of v_3 is 3, v_3 is connected by an edge to v_6 , therefore the degree of v_6 is 3

94.) Solution: (A)

Each subnet has 30 hosts

We need at least 5 host bits

For maximum number of subnets the subnet mask $24 + 3 = 27$

So 10.188.31.0/27 is the first address of the block.

95.) Solution: (D)

$10 \times 2\text{KB} + (2048/4) \times 2\text{KB} + (2048/4) \times (2048/4) \times 2\text{KB} = 20\text{K} + 1024\text{K} + 524288\text{K} = 525332\text{KB}$ (or 537939968B or 513.02MB etc.)

96.) Solution B 19

$\text{rec}(5) = \text{rec}(4) + 2 \times \text{rec}(3) + \text{rec}(2)$

$\text{rec}(4) = \text{rec}(3) + 2 \times \text{rec}(2) + \text{rec}(1)$

$\text{rec}(3) = \text{rec}(2) + 2 \times \text{rec}(1) + \text{rec}(0)$

$\text{rec}(2) = 2$

$\text{rec}(1) = 1$

$\text{rec}(0) = 0$

so $\text{rec}(5) = 19$

97.) Solution: (B)

$\text{foo}(0) \Rightarrow \text{inp}=1, \text{foo}(1), \text{print}(\text{inp})$

$\text{foo}(1) \Rightarrow \text{inp}=2, \text{foo}(2), \text{print}(\text{inp})$

$\text{foo}(2) \Rightarrow \text{inp}=3, \text{foo}(3), \text{print}(\text{inp})$

$\text{foo}(3) \Rightarrow \text{inp}=4, \text{foo}(4), \text{print}(\text{inp})$

$\text{foo}(4) \Rightarrow \text{inp}=5, \text{foo}(5), \text{print}(\text{inp})$

$\text{foo}(5) = \text{return}$

It will print 5 4 3 2 1

98.) Solution: A

B+ tree with order 3 means that at maximum we can have 3 block pointers in a node while minimum is 2. For getting the maximum number of nodes, we need to utilize the maximum possible block pointer as a node in next levels. So the tree will be like:

Level	Maximum nodes	Maximum block pointers
0	1 //root node	3
1	3	$3 \times 3 = 9$
2	9	$9 \times 3 = 27$
3	27	$27 \times 3 = 81$
4	81	$81 \times 3 = 243$
5	243	$243 \times 3 = 729$
Total nodes	364	

The height of the node is the number of edges on the longest path from the node to a leaf. So we have taken upto 5th level, so this way root to leaf, number of edges will be 5 and maximum number of nodes will be 364.

99.) SOLUTION D

The Boolean function $F(A, B, C, D)$ evaluates to 1 for minterms 1 and 9, and it evaluates to 0 for minterms 0 and 12. Thus, Option D) $F(A, B, C, D) = \Sigma(1, 3, 4, 6, 7, 8, 9, 11)$ is correct.

100.) SOLUTION A

The eigenvalues of the inverse matrix are equal to the reciprocal of the eigenvalues of the original matrix.

Eigenvectors of the matrix and its inverse are the same.